

MIS: Diagnosing Bottlenecks in Copy-Paste Augmentation for Agricultural Object Detection

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Abstract—Copy-paste augmentation achieves strong results for instance segmentation yet yields inconsistent or negative gains for bounding-box detection in agricultural scenes. We propose the Minimum Inference Score (MIS), a difficulty metric derived from a trained baseline detector that correlates strongly with detection failure modes—miss rate and false-positive rate—across objects and images. MIS partition analysis reveals whether hard instances concentrate in constrained regimes—driven by object size, domain composition, or both. When such concentration is detected, copy-paste fails to provide reliable gains across all tested selection strategies. Furthermore, by examining which difficulty groups dominate each MIS partition, practitioners can anticipate which selection strategies will produce the most harmful outcomes—all before any augmented training run. We validate both findings across three agricultural benchmarks, providing bidirectional validation across constrained and unconstrained datasets.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural object detection, data augmentation diagnostic

I. INTRODUCTION

Copy-paste augmentation composites cropped instances into training images to increase diversity [1], [2], yet yields inconsistent or negative gains for bounding-box detection in agricultural imagery, forcing costly multi-seed search without a principled stopping criterion. This failure persists across all tested selection strategies, suggesting a dataset-level bottleneck that no sampling policy can resolve.

We introduce the **Minimum Inference Score (MIS)**, a metric computed from a trained baseline detector that captures model failure patterns on a given dataset. MIS partition analysis enables two practical outputs without any augmented run: it determines whether copy-paste can supply useful training signal, by testing whether hard instances concentrate in constrained regimes—validated across MinneApple [3], GWHD 2021 [4], and WGISD [5]—and it predicts which selection strategies risk harmful outcomes, purely from partition composition.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ denote ground-truth boxes in image I . A trained detector produces predictions p with confidence c_p at $\tau_{\text{conf}}=0.001$ to retain near-miss candidates. For each $g \in G(I)$, let $\mathcal{P}(g)=\{p : \text{IoU}(p, g) \geq 0.5\}$. The *Object Difficulty Score* is

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset \\ \alpha + \beta, & \mathcal{P}(g) = \emptyset \end{cases} \quad (1)$$

with $\alpha=\beta=0.5$ and $u_p=\text{IoU}(p, g)$. Unmatched objects receive the maximum score ($\alpha+\beta=1$). The *Image Difficulty Score* is

$$S_{\text{img}}(I) = \frac{1}{|G(I)|} \sum_g S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I), \quad (2)$$

where $\text{FPrate}(I)$ counts unmatched predictions per image, giving $S_{\text{obj}} \in [0, 1]$ per object and S_{img} per image. By construction, S_{img} encodes both axes of detection failure (miss rate and FP rate); w_2 is set by grid search to balance their contributions. MIS is model-conditioned, reflecting structural properties only through detector behavior.

B. Augmentation Protocol

We train a YOLOv8s baseline and partition all training objects by S_{obj} and all training images by S_{img} into equal tertiles (Low, Medium, High). Four copy-paste conditions are evaluated: **Score High** pastes objects from the highest- S_{obj} tertile into the highest- S_{img} images; **Score Medium** and **Score Low** follow the same logic for their respective tertiles. **Random** selects both destination images and source objects without score-based filtering. This design isolates whether targeting specific score regimes affects outcome. Training data is expanded by 30% via unblended within image copy-paste [1]: 2–8 objects per image, scale $[0.9, 1.1]$, rotation $\pm 15^\circ$, $3\times$ maximum reuse; all other settings follow default YOLOv8 protocol.

III. RESULTS

A. MIS Reflects Detection Failure

Pearson correlations between S_{img} and miss rate on the validation set are $r=0.79, 0.81, 0.91$ on MinneApple, GWHD, and WGISD; correlations with false positive rate are $r=0.79, 0.81, 0.84$. These confirm that images with high S_{img} are genuinely harder to detect—the model misses more objects and produce more false positives on them—validating MIS as a reliable difficulty proxy.

B. MIS Identifies Dataset Failure Sources

Size (MinneApple). S_{obj} strongly correlates with object size (Spearman $r_s=-0.82$): 64.4% of the top-difficulty tertile falls in *Very Small* instances ($\leq 400 \text{ px}^2$, low AP), indicating the model struggles primarily with small objects.

Domain (GWHD 2021). Per-domain median MIS strongly correlates with domain AP (Pearson $r=-0.90$; Fig. 1): MIS assigns higher scores to harder acquisition domains, purely from detector behavior and without access to domain labels.

TABLE I
MINNEAPPLE: S_{obj} CAPTURES SIZE-LEVEL DIFFICULTY.

Category	AP	Mean S_{obj}	Top tertile
Very Small ($\leq 400\text{px}^2$)	0.071	0.152	64.4%
Small (400–1024 px^2)	0.270	0.076	33.9%
Medium ($> 1024\text{px}^2$)	0.493	0.053	1.7%

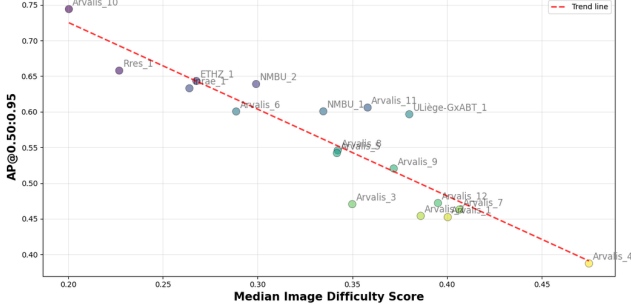


Fig. 1. GWHD 2021: per-domain AP vs. median image difficulty score.

C. Application: Copy-Paste Diagnostic and Strategy Risk

MinneApple (resolution ceiling). We define *resolution ceiling* as a size below which an object’s footprint on the detector’s feature maps is too small to localize reliably, no matter how many training examples exist. We call this a *structural constraint*: it is a limit of representation capacity, not of data quantity. MIS partition analysis shows 64.4% of high-difficulty objects fall in the Very Small category ($\leq 400\text{px}^2$, $\text{AP} = 0.071$). This signals a resolution ceiling: pasting additional objects provides no useful learning signal because the model already fails on them structurally, not from lack of examples. This is confirmed: all copy-paste conditions yield negative ΔAP (Table II), regardless of selection strategy, with no condition reversing the outcome direction.

GWHD 2021 (domain difficulty and strategy risk). MIS partitioning reveals near-complete domain separation: low-difficulty images are 66% ETHZ_1/Rres_1 ($\text{AP} \approx 0.65$), while high-difficulty images are 78% Arvalis variants ($\text{AP} \approx 0.46$). Since difficulty stems from each domain’s image-level properties rather than a lack of instances, copy-paste cannot help: it adds instances but cannot change how a domain looks to the detector. This is confirmed: all conditions yield negative ΔAP . Partition composition further predicts per-strategy risk from the domains each strategy sources from: Score Low draws exclusively from high-AP domains, flooding training with redundant patterns (consistent drop, -0.034 , $\sigma=0.005$); Score High draws from hard Arvalis domains, introducing high-variance signal the model cannot reliably exploit (seed instability, -0.006 , $\sigma=0.040$); Random and Medium source from mixed partitions, giving moderate, more stable drops (-0.015 – -0.017 , $\sigma=0.023/0.021$).

WGISD (positive control). MIS raises no flag: difficulty is distributed across size categories without concentration, and no domain separation is present, signaling that copy-paste can introduce useful learning signal. This is confirmed: all conditions yield positive ΔAP (Table II), with Score Medium achieving the best result.

TABLE II
COPY-PASTE OUTCOMES (3-SEED MEAN $\pm \sigma$). BOTTLENECK: FAILURE SOURCE IDENTIFIED BY MIS.

Dataset	Bottleneck	Base AP	Best Δ	Worst Δ
MinneApple	Size	0.353 ± 0.005	-0.003 (Hi)	-0.008 (Rnd)
GWHD 2021	Domain	0.229 ± 0.013	-0.006^* (Hi)	-0.034 (Lo)
WGISD	—	0.480 ± 0.007	$+0.009$ (Med)	$+0.001$ (Hi)

*Score High on GWHD: $\sigma=0.040$, high seed variance.

D. Limitations

Two aspects remain unvalidated: w_2 in Eq. (2) was fixed by grid search on one configuration, so its effect on the tertile partitions and diagnostic conclusions is untested; and all results use a single detector, YOLOv8s, so transfer to other architectures is untested. Both are left to future work, via sensitivity analysis and cross-architecture validation.

IV. CONCLUSION

Copy-paste augmentation fails when hard instances concentrate in structurally constrained regimes—below the detector’s resolution range or within acquisition-defined domains that instance-level augmentation cannot bridge. MIS makes these constraints visible before any augmented run: on MinneApple and GWHD 2021, partition analysis correctly predicted failure across all four selection strategies; on WGISD, the absence of such concentration correctly predicted positive gains across all conditions—providing bidirectional validation that MIS functions as a diagnostic, not merely a failure detector. On domain-structured datasets, MIS partition composition delivers a second output: it identifies risky selection strategies from partition composition alone, before any augmented run. Across our experiments, 24 augmented runs on MIS-flagged datasets yielded zero reliable gains, a compute cost MIS avoids entirely. These findings hold for the single weighting and detector validated here (Section III-D); future work includes guiding instance-level selection when constraints are absent, and pairing MIS with generative methods for hard regimes copy-paste cannot reach.

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